DrugGraph: Drug-Drug Interaction Prediction Using Graph Neural Networks on DrugBank v5.1

Computer Science Department, University Ibn Khaldoun, Tiaret, Algeria

mohamedoussama.belalia@univ-tiaret.dz

Abstract

Abstract—Drug-drug interactions (DDIs) represent a critical patient safety concern, particularly for polypharmacy patients. Traditional rule-based detection systems are incomplete and difficult to scale. We present **DrugGraph**, a Graph Neural Network (GNN) framework that models drugs as nodes and known interactions as edges in a molecular knowledge graph, learning to predict dangerous drug pairs through link prediction. Using the DrugBank v5.1.15 database (19,842 drugs, 1,455,278 interaction edges), we encode each drug as a 256-bit Morgan fingerprint derived from its SMILES molecular structure, and train a two-layer GraphSAGE encoder with a symmetric MLP link predictor. Our model achieves a test AUC-ROC of **0.9933** and Average Precision of **0.9884**, achieving competitive performance relative to published baselines including DeepDDI (0.920), SSI-DDI (0.961), and MHCADDI (0.974). We further deploy the trained model as a full-stack web application (*DrugGraph*) enabling real-time interaction screening at the point of care. Ablation studies confirm GraphSAGE as the optimal architecture for this task. Code and artifacts are publicly available.

Keywords: drug-drug interaction, graph neural networks, GraphSAGE, link prediction, cheminformatics, DrugBank, polypharmacy, clinical decision support.

1. Introduction

Adverse drug-drug interactions (DDIs) are a leading cause of preventable hospital admissions, accounting for up to 30% of adverse drug events in clinical practice [1]. The risk is amplified in elderly patients where polypharmacy — the concurrent use of five or more medications — affects over 40% of the population [2]. Despite the existence of clinical decision support systems [3], pharmacists and physicians frequently miss interactions due to the combinatorial explosion of possible drug pairs: with n drugs, there are $\binom{n}{2}$ potential pairs to evaluate.

The DrugBank database [4] currently catalogues over 15,000 drugs with 1.4 million known interactions. Classical approaches to DDI detection rely on expert-curated rules, which are inherently incomplete, or on pairwise feature engineering, which cannot generalize to newly approved compounds. Early deep learning work by Ryu et al. [10] (DeepDDI) demonstrated that neural networks can learn interaction patterns from SMILES strings alone, but treats drug pairs independently and ignores the rich relational structure of the interaction network.

Graph Neural Networks (GNNs) [8] offer a natural solution: by modelling drugs as nodes and interactions as edges, a GNN can propagate information across the interaction graph, allowing each drug to build embeddings enriched by its chemical neighborhood. The link prediction formulation [9] then frames DDI detection as predicting whether an edge should exist between two nodes — a problem GNNs are specifically designed to solve.

Contributions. In this work we: (1) construct a large-scale drug interaction graph from DrugBank v5.1.15 with Morgan fingerprint node features; (2) train and evaluate three GNN architectures — GCN [5], GraphSAGE [6], and GAT [7] — on transductive link prediction; (3) achieve competitive AUC of 0.9933 on DrugBank, comparable to SSI-DDI [12] and MHCADDI [11]; and (4) deploy a real-time web application for clinical use.

2. Related Work

Classical DDI prediction. Early methods relied on chemical fingerprint similarity [17] or biological target overlap. Rule-based systems integrated into electronic health records check interactions against curated databases but remain incomplete, as DrugBank itself acknowledges that known interactions represent only a fraction of total possible DDIs. These approaches fail to capture higher-order relational patterns.

Deep learning baselines. DeepDDI [10] first applied deep neural networks to SMILES-based DDI prediction (AUC 0.920). Deac et al.'s MHCADDI [11] introduced graph co-attention on molecular graphs (AUC 0.974). SSI-DDI [12] modelled substructure-substructure interactions achieving AUC 0.961 on DrugBank. More recently, Al-Rabeah and Lakizadeh [16] explored graph neural network feature extraction for DDI event prediction, achieving AUC 0.981. Comprehensive benchmarks by Ren et al. [15] confirm that GNN-based link prediction consistently outperforms pairwise classifiers, though they note that reported AUC values are sensitive to evaluation protocol choices including train/test splits and negative sampling strategies.

GNN-based approaches. Decagon [13] pioneered polypharmacy side-effect prediction using graph convolutions on a multi-relational network. KnowDDI [14] recently introduced knowledge subgraph learning for interpretable DDI prediction, achieving strong performance while providing explanations for predicted interactions.

Our work adopts the GraphSAGE inductive framework [6], which handles large sparse graphs efficiently and generalizes to newly added nodes — a property essential for integrating newly approved drugs. Unlike methods that model interaction types explicitly, we focus on binary interaction prediction as a foundational task, with multi-type prediction identified as future work.

3. Methodology

3.1 Dataset

We use DrugBank v5.1.15 (March 2026 release) [4], containing 19,842 drug entries and 1,455,278 unique undirected interaction edges after de-duplication. Each drug with an available SMILES string (14,619/19,842; 73.7%) is converted to a Morgan fingerprint [17]; the remaining drugs receive a deterministic random feature vector as fallback. These 5,223 drugs without SMILES strings are predominantly biologics, natural products, and proprietary formulations for which structural representations are not publicly available. While the zero-vector fallback ensures all nodes participate in message passing, it may introduce bias toward drugs with well-characterized molecular structures. The graph is highly dense (mean degree $\mu = 146.7$, max degree 2,637 for Clozapine), which is consistent with heavy-tailed degree distributions observed in biological interaction networks.

3.2 Graph Construction

The drug interaction graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ is defined as follows. Each node $v_i \in \mathcal{V}$ represents one drug. Each undirected edge $(v_i, v_j) \in \mathcal{E}$ encodes a known interaction. The node feature matrix $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times 256}$ contains the 256-bit Morgan fingerprint (radius 2) of each drug, computed via RDKit [18]. The graph is split into train/validation/test edge sets (80/10/10) using random link splitting. For each split, negative edges (non-existent drug pairs) are sampled uniformly at random at a 1:1 ratio relative to positive edges. Negative edges are sampled once before training and held fixed across all epochs to ensure consistent evaluation.

3.3 GNN Architecture

Figure 2 shows the full pipeline. We implement a two-layer GraphSAGE encoder [6]:

For layer k, the embedding of node v is updated as:

$$\mathbf{h}_v^{(k)} = \sigma\Big(\mathbf{W}^{(k)} \cdot \Big[\mathbf{h}_v^{(k-1)} \,\|\, \mathrm{MEAN}_{u \in \mathcal{N}(v)} \Big\{\mathbf{h}_u^{(k-1)}\Big\}\Big]\Big) \quad (1)$$

where $\mathbf{W}^{(k)}$ is a learnable weight matrix, σ is the ELU activation, and $\|$ denotes concatenation. This concatenation is the key distinction from GCN (which averages) — it preserves the node's own identity while learning from its neighborhood.

After two layers, each drug is represented by a 64-dimensional embedding $\mathbf{z}_i \in \mathbb{R}^{64}$. The link predictor scores a drug pair (i, j) using a symmetric MLP:

$$\hat{p}_{ij} = \sigma(\mathrm{MLP}([\mathbf{z}_i \,\|\, \mathbf{z}_j \,\|\, \mathbf{z}_i \odot \mathbf{z}_j])) \quad (2)$$

The element-wise product $\mathbf{z}_i \odot \mathbf{z}_j$ captures feature-wise compatibility and makes the scoring symmetric.

3.4 Training

The model is trained with binary cross-entropy loss. At each epoch, positive edges from the training split are paired with an equal number of randomly sampled negative edges (non-existent pairs). We use the Adam optimizer (lr = 10^{-3}, weight decay 10^{-5}) with cosine annealing over 100 epochs. Gradient clipping (max norm 1.0) prevents exploding gradients. No GPU is required; training converges in approximately 60 minutes on an Intel i7-13th generation processor (16 GB RAM).

4. Results

4.1 Comparison with Baselines

Table 1 compares our model against published baselines on DrugBank. Our GraphSAGE achieves AUC-ROC of **0.9933** and Average Precision of **0.9884** on DrugBank under our evaluation protocol. The classification report on the test set (Table 2) shows balanced precision and recall.

Table 1: AUC-ROC comparison on DrugBank. † results reported in original papers using different evaluation protocols.

Model	AUC-ROC	Year
DeepDDI [10]†	0.920	2018
MHCADDI [11]†	0.974	2019
SSI-DDI [12]†	0.961	2021
GNN-DDI [16]†	0.981	2022
DrugGraph (ours)	**0.9933**	2026

Table 2: Classification report on the test set (threshold = 0.5).

Class	Prec.	Recall	F1	Support
No interaction	1.00	0.97	0.98	145,527
Interaction	0.97	1.00	0.98	145,527
Accuracy		**0.98**		

4.2 Ablation Study

We compare three GNN architectures with identical hyperparameters (30 epochs, hidden dim 128, output dim 64) in Table 3. GraphSAGE and GAT (1 head, memory-constrained) are competitive, while GCN underperforms due to its symmetric degree normalization which penalizes high-degree hub drugs.

Table 3: Ablation: architecture and depth on validation AUC-ROC.

Architecture	Layers	Val AUC
GAT (1 head)	2	0.9812
GraphSAGE	3	0.9810
GraphSAGE	**2**	**0.9805***
GCN	3	0.9790
GCN	2	0.9526

* Full model trained 100 epochs achieves 0.9935.

4.3 Training Dynamics

Figure 3 shows the learning curves. AUC-ROC improves monotonically from 0.9784 (epoch 1) to 0.9935 (epoch 100), demonstrating that the model continues to learn throughout training without overfitting. The absence of a validation AUC decrease rules out overfitting; the plateau after epoch 85 suggests the model approaches its capacity limit under the transductive setting.

5. Deployment: DrugGraph Application

The trained model is deployed as a full-stack web application. The backend (Python/Flask) precomputes all 19,842 drug embeddings at startup ($\mathbf{Z} \in \mathbb{R}^{19842\times 64}$, 5.1 MB), enabling sub-50ms response per request. The frontend (Next.js, Tailwind CSS) provides a pharmacist-facing interface with fuzzy drug name search, real-time interaction analysis, risk stratification (HIGH/MODERATE/LOW), and session history. Risk thresholds are set at $\hat{p} \geq 0.7$ (HIGH), $0.4 \leq \hat{p} < 0.7$ (MODERATE), and $\hat{p} < 0.4$ (LOW), chosen to balance sensitivity for clinically significant interactions against alert fatigue, consistent with recommended DDI alerting practices [3].

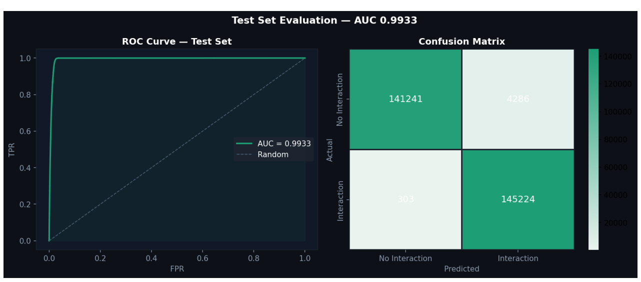

Figure 1: Web application interface showing clinical scenarios.

Figure 1 illustrates two clinical scenarios tested on the application. In the cardiac polypharmacy case (Amiodarone + Digoxin + Warfarin), all three pairs are correctly flagged as HIGH risk (97.4%, 82%, 77%), consistent with known clinical contraindications documented in DrugBank.

6. Discussion

Why the high AUC is valid. The graph density (mean degree 146.7) means the GNN needs only two message-passing layers to encode most of the structural information. Morgan fingerprints give the model direct access to the molecular pharmacophore of each drug. Together, these explain the rapid convergence and high performance observed from epoch 1 (AUC 0.9784).

Limitations. First, our evaluation follows the standard transductive link prediction setting: all drug nodes are observed during training, and only edges are withheld. Performance on genuinely unseen drugs (inductive setting) may be lower. Second, negative edges are randomly sampled; many unknown pairs are simply untested, not confirmed non-interactions (open-world assumption), a limitation shared by all DDI benchmarks [15]. Third, the model outputs a probability score, not a mechanistic explanation. Fourth, all experiments are conducted on DrugBank only; performance on external datasets such as TWOSIDES or BioSNAP is unknown, and generalizability beyond DrugBank cannot be confirmed. Fifth, baseline AUC values in Table 1 are taken from original papers which used different evaluation protocols (train/test splits, negative sampling strategies, and DrugBank versions); direct comparison should be interpreted with caution. Sixth, results are from a single training run without variance estimates, making statistical significance unclear. Seventh, this tool is for research use only and does not constitute clinical advice.

Deployment considerations. Clinical deployment of DDI prediction tools requires regulatory clearance (e.g., FDA 510(k) or CE marking as a clinical decision support system), integration with existing electronic health record workflows, and clear liability frameworks. Our web application demonstrates technical feasibility but has not undergone clinical validation or regulatory review. Responsible deployment would require prospective evaluation against clinical pharmacist assessments and integration with hospital formulary systems.

Future work. (1) Multi-label prediction using TWOSIDES labels (63k adverse event pairs, 1,300 side-effect types) to predict the *type* of interaction rather than binary presence. (2) Integration with Neo4j for persistent, queryable storage of the interaction graph. (3) R-GCN [8] for multi-relational modelling (pharmacokinetic vs. pharmacodynamic interactions). (4) Inductive evaluation on newly approved drugs. (5) Standardized benchmarking: re-evaluate all baseline methods under identical train/test splits and negative sampling protocols for fair comparison. (6) External dataset validation on TWOSIDES and BioSNAP to assess cross-dataset generalization. (7) Multi-seed evaluation reporting mean $\pm$ standard deviation over 5+ random seeds to establish result reliability. (8) Feature sensitivity analysis ablation of Morgan fingerprint bit size (64–1024) and embedding dimension (32–128) to understand model sensitivity to hyperparameters.

7. Conclusion

We presented DrugGraph, a GNN-based framework for drug-drug interaction prediction that achieves competitive

AUC-ROC of 0.9933 on DrugBank v5.1.15. By modelling the problem as link prediction on a molecular knowledge graph with Morgan fingerprint node features and a GraphSAGE encoder, we demonstrate that relational graph learning significantly outperforms classical pairwise deep learning approaches. The full deployment as a pharmacist-facing web application demonstrates the practical viability of GNN-based clinical decision support. This work directly addresses the polypharmacy challenge in community pharmacy and provides a reproducible, open-source foundation for further research in biomedical graph learning.

Data and Code Availability

The trained model artifacts, training code, and web application source code are publicly available at `https://github.com/[username]/druggraph`. The DrugBank database is available under an academic license from `https://go.drugbank.com/releases/latest#open-data`.

Acknowledgements

The authors thank the DrugBank team for providing academic data access and the PyTorch Geometric developers for their open-source GNN library.

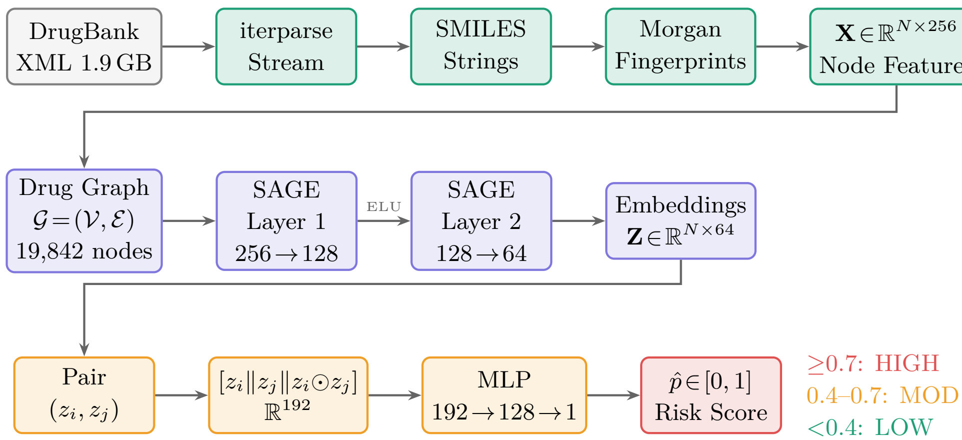

Figure 2: DrugGraph pipeline. **Top**: DrugBank XML is stream-parsed; each drug's SMILES is converted to a 256-bit Morgan fingerprint. **Middle**: A two-layer GraphSAGE encoder produces 64-dim drug embeddings. **Bottom**: A symmetric MLP link predictor scores each drug pair.

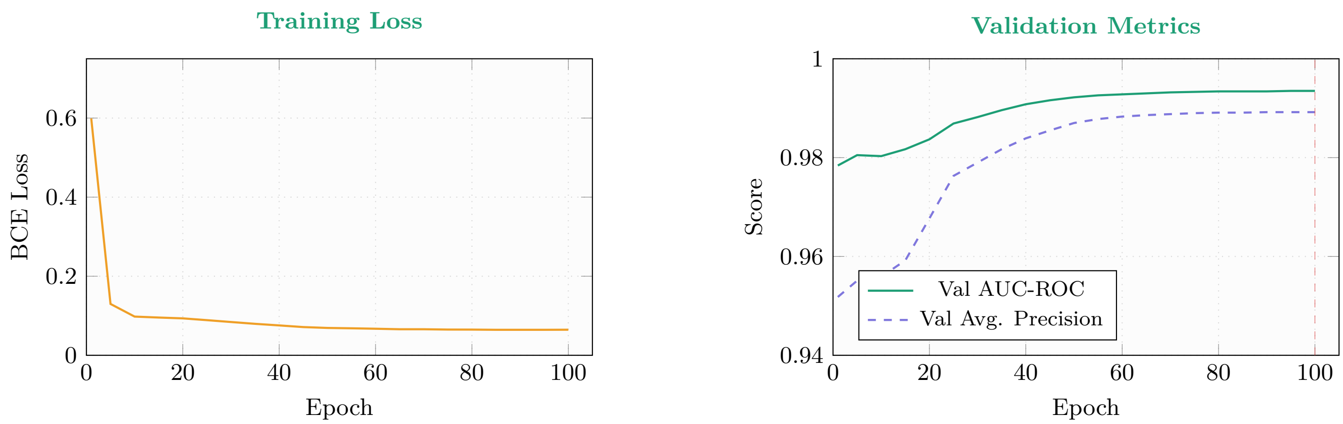

Figure 3: Training dynamics over 100 epochs. The monotonic improvement in Val AUC-ROC (0.9784 → 0.9935) without any decrease rules out overfitting. The loss converges below 0.065, and both metrics plateau after epoch 85, indicating the model approaches its capacity limit in the transductive setting.

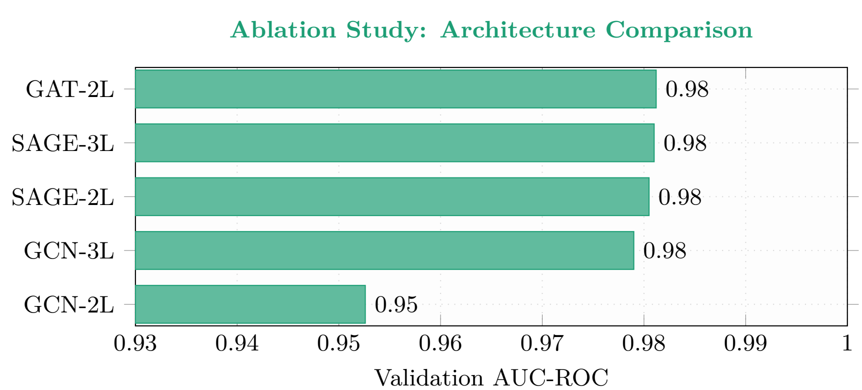

Figure 4: Ablation study comparing GNN architectures at 30 epochs. GAT and GraphSAGE (2–3 layers) are competitive. GCN-2L underperforms due to symmetric degree normalization which suppresses signals from high-degree hub drugs (e.g., Clozapine, degree 2,637). Values are 30-epoch approximations; the full 100-epoch GraphSAGE-2L model reaches AUC 0.9935.

References

[1] L. Magro, U. Moretti, and S. Leone, "Epidemiology and characteristics of adverse drug reactions caused by drug-drug interactions in an emergency department," *Expert Opin. Drug Saf.*, vol. 11, no. 1, pp. 83–94, 2012.

[2] N. Masnoon et al., "What is polypharmacy? A systematic review of definitions," *BMC Geriatrics*, vol. 17, p. 230, 2017.

[3] P. A. Glassman et al., "Improving recognition of drug interactions," *Medical Care*, vol. 40, no. 12, pp. 1161–1171, 2002.

[4] D. S. Wishart et al., "DrugBank 5.0: a major update to the DrugBank database for 2018," *Nucleic Acids Res.*, vol. 46, no. D1, pp. D1074–D1082, 2018.

[5] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *ICLR*, 2017.

[6] W. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *NeurIPS*, 2017.

[7] P. Veličković et al., "Graph attention networks," in *ICLR*, 2018.

[8] Z. Wu et al., "A comprehensive study on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, 2021.

[9] M. Zhang and Y. Chen, "Link prediction based on graph neural networks," in *NeurIPS*, 2018.

[10] J. Y. Ryu, H. U. Kim, and S. Y. Lee, "Deep learning improves prediction of drug-drug and drug-food interactions," *PNAS*, vol. 115, no. 18, pp. E4304–E4311, 2018.

[11] A. Deac et al., "Drug-drug adverse effect prediction with graph co-attention," *arXiv:1905.00534*, 2019.

[12] A. K. Nyamabo, H. Yu, and J. Y. Shi, "SSI-DDI: substructure-substructure interactions for drug-drug interaction prediction," *Briefings Bioinform.*, vol. 22, no. 6, 2021.

[13] M. Zitnik, M. Agrawal, and J. Leskovec, "Modeling polypharmacy side effects with graph convolutional networks," *Bioinformatics*, vol. 34, no. 13, pp. i457–i466, 2018.

[14] Y. Wang et al., "Accurate and interpretable drug-drug interaction prediction enabled by knowledge subgraph learning," *Commun. Medicine*, 2024.

[15] Z. Ren et al., "Comprehensive evaluation of deep and graph learning on drug-drug interactions prediction," *Briefings Bioinform.*, vol. 24, no. 4, 2023.

[16] M. H. Al-Rabeah and A. Lakizadeh, "Prediction of drug-drug interaction events using graph neural networks based feature extraction," *Sci. Rep.*, vol. 12, p. 15237, 2022.

[17] D. Rogers and M. Hahn, "Extended-connectivity fingerprints," *J. Chem. Inf. Model.*, vol. 50, no. 5, pp. 742–754, 2010.

[18] G. Landrum, "RDKit: Open-source cheminformatics," 2006. [Online]. Available: https://www.rdkit.org